MAE 3405
FLIGHT DYNAMICS

The University of Texas at Arlington

Project 2

The Projectile Motion Problem, Revisited

At a coordinated universal time of UTC = 2025:12:13:14:15:16, the students of MAE 3405 (Flight Dynamics) launch a prototype version of the "Perigalacticon" projectile from the middle of the 50-yard line of the AT&T Stadium football field (ϕ_{gs} = 32° 44' 52'' N, λ_{gs} = 97° 5' 34'' W, h_{gs} = 185 m). The projectile's speed, azimuth, elevation, and roll angle with respect to the launch site at that instant are 827 m/s, 107°, 45°, and 0°, respectively. Develop a six degree of freedom simulation of the projectile's trajectory while considering the Magnus force, the drag force, and the acceleration due to gravity experienced by the projectile for a period equal to the time that it takes the shell to impact the initial altitude above mean equator after launch.

Projectile Characteristics

The projectile has the following characteristics:

- m_{cg} = 35 kg (mass)
- c_l = 0.173 (Magnus coefficient)
- c_d = 0.820 (drag coefficient)
- a = 6006.25π mm^2 (reference area)
- I_{xx} = 0.0630656250 kg-m^2
- I_{yy} = 1.3440328125 kg-m^2
- I_{zz} = 1.3440328125 kg-m^2
- $I_{xy} = I_{xz} = I_{yz}$ = 0.1579431538 kg-m^2

Forces

The forces that should be considered in this analysis should be the Magnus force, the drag force, and the projectile's weight.

$$\vec{F}_m = \frac{1}{2} c_l \rho a v_\infty^2 \left(\frac{\vec{\omega}_{cg} \times \vec{v}_\infty}{|\vec{\omega}_{cg} \times \vec{v}_\infty|} \right) \tag{1}$$

$$\vec{F}_d = -\frac{1}{2} c_d \rho a v_\infty \vec{v}_\infty \tag{2}$$

$$\vec{g} = -\frac{u}{r_{cg/\oplus}^3} \vec{r}_{cg/\oplus} \tag{3}$$

Moments

The moments that should be considered in this analysis should be the moment due to the Magnus force and the moment due to the drag force.

$$\vec{M}_m^{bod} = \vec{r}_{cp/cg}^{bod} \times \vec{F}_m^{bod} \tag{4}$$

$$\vec{M}_d^{bod} = \vec{r}_{cp/cg}^{bod} \times \vec{F}_d^{bod} \tag{5}$$

Plots

The following plots should be submitted:

1. A 3 x 1 subplot configuration containing the following subplots:
   - Altitude above mean equator vs. Time.
     - The title should read "Altitude" with a bold, 16-point, Arial font.
     - The x-label should read "Time (s)" with a bold, 12-point, Arial font.
     - The y-label should read "Altitude (km)" with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Linear range with respect to the ground station vs. Time.
     - The title should read "Linear" with a bold, 16-point, Arial font.
     - The x-label should read "Time (s)" with a bold, 12-point, Arial font.
     - The y-label should read "Range (km)" with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Haversine range with respect to the ground station vs. Time.
     - The title should read "Haversine" with a bold, 16-point, Arial font.
     - The x-label should read "Time (s)" with a bold, 12-point, Arial font.
     - The y-label should read "Range (km)" with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.

2. A 3 x 1 subplot configuration containing the following subplots:
   - Relative speed with respect to the ground station vs. Time.
     - The title should read "Relative" with a bold, 16-point, Arial font.
     - The x-label should read "Time (s)" with a bold, 12-point, Arial font.
     - The y-label should read "Speed (m/s)" with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Ground speed vs. Time.
     - The title should read "Ground" with a bold, 16-point, Arial font.
     - The x-label should read "Time (s)" with a bold, 12-point, Arial font.
     - The y-label should read "Speed (m/s)" with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Dynamic pressure vs. Time.
     - The title should read "Dynamic" with a bold, 16-point, Arial font.
     - The x-label should read "Time (s)" with a bold, 12-point, Arial font.
     - The y-label should read "Pressure (kPa)" with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.

3. A 3 x 1 subplot configuration containing the following subplots:
   - Eastern Displacement vs. Time
     - The title should read "East" with a bold, 16-point, Arial font.
     - The x-label should read "Time (s)" with a bold, 12-point, Arial font.
     - The y-label should read "Displacement (km)" with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Northern Displacement vs. Time
     - The title should read "North" with a bold, 16-point, Arial font.
     - The x-label should read "Time (s)" with a bold, 12-point, Arial font.
     - The y-label should read "Displacement (km)" with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Zenith Displacement vs. Time
     - The title should read "Zenith" with a bold, 16-point, Arial font.
     - The x-label should read "Time (s)" with a bold, 12-point, Arial font.
     - The y-label should read "Displacement (km)" with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.

4. A 3 x 1 subplot configuration containing the following subplots:
   - Eastern Speed vs. Time
     - The title should read “East” with a bold, 16-point, Arial font.
     - The x-label should read “Time (s)” with a bold, 12-point, Arial font.
     - The y-label should read “Speed (m/s)” with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Northern Speed vs. Time
     - The title should read “North” with a bold, 16-point, Arial font.
     - The x-label should read “Time (s)” with a bold, 12-point, Arial font.
     - The y-label should read “Speed (m/s)” with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Zenith Speed vs. Time
     - The title should read “Zenith” with a bold, 16-point, Arial font.
     - The x-label should read “Time (s)” with a bold, 12-point, Arial font.
     - The y-label should read “Speed (m/s)” with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.

5. A 2 x 2 subplot configuration containing the following subplots:
   - X-component of inertial acceleration in body coordinates vs. Time
     - The title should read “Body-X” with a bold, 16-point, Arial font.
     - The x-label should read “Time (s)” with a bold, 12-point, Arial font.
     - The y-label should read “Acceleration (g’s)” with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Y-component of inertial acceleration in body coordinates vs. Time
     - The title should read “Body-Y” with a bold, 16-point, Arial font.
     - The x-label should read “Time (s)” with a bold, 12-point, Arial font.
     - The y-label should read “Acceleration (g’s)” with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Z-component of inertial acceleration in body coordinates vs. Time
     - The title should read “Body-Z” with a bold, 16-point, Arial font.
     - The x-label should read “Time (s)” with a bold, 12-point, Arial font.
     - The y-label should read “Acceleration (g’s)” with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Inertial acceleration vs. Time.
     - The title should read “Inertial” with a bold, 16-point, Arial font.
     - The x-label should read “Time (s)” with a bold, 12-point, Arial font.
     - The y-label should read “Acceleration (g’s)” with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.

6. A 2 x 1 subplot configuration containing the following subplots:
   - Azimuth vs. Time
     - The title should read “Azimuth” with a bold, 16-point, Arial font.
     - The x-label should read “Time (s)” with a bold, 12-point, Arial font.
     - The y-label should read “Azimuth (°)” with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Elevation vs. Time
     - The title should read “Elevation” with a bold, 16-point, Arial font.
     - The x-label should read “Time (s)” with a bold, 12-point, Arial font.
     - The y-label should read “Elevation (°)” with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.

7. A 3 x 1 subplot configuration containing the following subplots:
   - Roll angle vs. Time
     - The title should read “Roll” with a bold, 16-point, Arial font.
     - The x-label should read “Time (s)” with a bold, 12-point, Arial font.
     - The y-label should read “Roll (°)” with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Pitch angle vs. Time
     - The title should read “Pitch” with a bold, 16-point, Arial font.
     - The x-label should read “Time (s)” with a bold, 12-point, Arial font.
     - The y-label should read “Pitch (°)” with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Yaw angle vs. Time
     - The title should read “Yaw” with a bold, 16-point, Arial font.
     - The x-label should read “Time (s)” with a bold, 12-point, Arial font.
     - The y-label should read “Yaw (°)” with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.

8. A 2 x 2 subplot configuration containing the following subplots:
   - X-component of the rotational velocity with respect to the center of gravity vs. Time
     - The title should read “Body-X” with a bold, 16-point, Arial font.
     - The x-label should read “Time (s)” with a bold, 12-point, Arial font.
     - The y-label should read “ω_x (°/s)” with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Y-component of the rotational velocity with respect to the center of gravity vs. Time
     - The title should read “Body-Y” with a bold, 16-point, Arial font.
     - The x-label should read “Time (s)” with a bold, 12-point, Arial font.
     - The y-label should read “ω_y (°/s)” with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Z-component of the rotational velocity with respect to the center of gravity vs. Time
     - The title should read “Body-Z” with a bold, 16-point, Arial font.
     - The x-label should read “Time (s)” with a bold, 12-point, Arial font.
     - The y-label should read “ω_z (°/s)” with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Rotational speed with respect to the center of gravity vs. Time.
     - The title should read “Total” with a bold, 16-point, Arial font.
     - The x-label should read “Time (s)” with a bold, 12-point, Arial font.
     - The y-label should read “ω (°/s)” with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.

9. A 2 x 2 subplot configuration containing the following subplots:
   - First quaternion component vs. Time
     - The title should read "Q_1" with a bold, 16-point, Arial font.
     - The x-label should read "Time (s)" with a bold, 12-point, Arial font.
     - The y-label should read "Q_1" with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Second quaternion component vs. Time
     - The title should read "Q_2" with a bold, 16-point, Arial font.
     - The x-label should read "Time (s)" with a bold, 12-point, Arial font.
     - The y-label should read "Q_2" with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Third quaternion component vs. Time
     - The title should read "Q_3" with a bold, 16-point, Arial font.
     - The x-label should read "Time (s)" with a bold, 12-point, Arial font.
     - The y-label should read "Q_3" with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.
   - Fourth quaternion component vs. Time
     - The title should read "Q_4" with a bold, 16-point, Arial font.
     - The x-label should read "Time (s)" with a bold, 12-point, Arial font.
     - The y-label should read "Q_4" with a bold, 12-point, Arial font.
     - The x- and y-axis tick mark labels should be typed in bold, 8-point, Arial fonts.

Scenario

In this scenario, assume that the position of the center of pressure with respect to the center of gravity during the flight is:

$$\vec{r}_{cp/cg}^{\,bod} = -\hat{x}^{bod}\ \text{mm} \tag{6}$$

Deliverables

The deliverables for this assignment shall be the nine plots submitted as portable network graphics (.png) files in a compressed folder named: "LastName_FirstName_StudentNumber_MAE3405_P2.zip". The nine figures shall be titled:

- Displacement.png
- Speed.png
- Position.png
- Velocity.png
- Acceleration.png
- AzimuthElevation.png
- RollPitchYaw.png
- AngularRates.png
- Quaternion.png